E6 | *fx* | =D6-SUMPRODUCT(G$14:G$19,$L$14:$L$19)

Rows 1 and 2: the title and the note on the model

3	Growth in volume (Assumptions)	40.0%							
4									
5	**Work center**	**Practical capacity**	**Load of the base year's sales**	**Load at growth, nothing bought**	**Hours used after buying**	**Slack**	**Status**		
6	Cutting	20,524	11,377	15,927	15,872	4,653	Not binding		
7	Assembly	44,142	31,542	44,159	44,052	90	Not binding		
8	Finishing and Test	24,187	17,323	24,252	24,187	0	**Binding**		
9	Packing	11,047	7,000	9,800	9,776	1,271	Not binding		
10	Quality Assurance	3,166	309	433	433	2,733	Not binding		
11									
12									
13	**Family**	**Base-year units**	**Units at growth (the most that can be bought)**	**Delivered quote**	**Relevant cost, crew kept**	**Added cost per unit bought**	**Cutting hours per unit**	**Units bought (variable cells)**	**Added cost**
14	FUR-BNH	1,454.9	2,036.9	334.75	332.16	2.59	0.3677	150.6	390.59
15	FUR-NGT	3,198.9	4,478.4	278.10	243.27	34.83	0.3612	0.0	0.00
16	FUR-SDB	3,780.0	5,291.9	334.75	267.98	66.77	0.3838	0.0	0.00
17	FUR-TBL	4,984.1	6,977.8	324.45	280.42	44.03	0.4181	0.0	0.00
18	LGT-SCN	4,591.4	6,428.0	185.40	143.68	41.72	0.0000	0.0	0.00
19	TXT-PIL	673.3	942.6	226.60	138.64	87.96	0.3057	0.0	0.00
20									
21	Total added cost (Solver's objective: minimize)	390.59							

Rows 22 to 32: the binding work centers, the families bought, the checks, and the plan the supplier would accept

Variable cells, L14:L19 | Constraints: E6:E10 not above B6:B10, L14:L19 not above C14:C19 | Objective, B21

The binding work center

Capacity | **Growth 40%** | Growth 40% Sensitivity

The model of Requirement 7's growth test at 40%: minimize the total added cost, B21, by changing the units bought, L14:L19, with the hours used after buying (E6:E10) not above the practical capacity (B6:B10) and the units bought not above the units at growth (C14:C19). Finishing and Test binds: its slack is 0 and an hour of its capacity is worth 6.01 (column I, hidden). The values in L14:L19 are the optimum computed by the simplex method and written when the workbook was built; Solver itself was not run. The formula bar shows the hours used after buying at the first work center. Columns H to K and N to R are hidden here: the shadow price, the hours per unit at each work center and the plan the supplier would accept.